

ISHANI MONDAL
PhD Student | UMD College Park, USA

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RESEARCH INTERESTS

My research interests primarily revolve around evaluating and advancing the capabilities of LLM agents in two different directions :

- 1) **Human-AI Interaction and Alignment.** This area focuses on improving the interaction between humans and AI systems through personalized responses and context-aware adaptations :
- a) [Personalized Presentation Generation using RLHF (EACL 2024) [🔗]],
 - b) [Adaptive Diagram Generation (EMNLP 2024) [🔗]],
 - c) [Adaptive IE (COLING 2025) [🔗]],
 - d) [Multi-Agent Collaboration to Facilitate Human-Like Design Editing(Under Submission)]
 - e) [Exploring the Factors of Personalization from In-situ Human-AI Conversations (EMNLP 2025)]
2. **LLM Transparency and Evaluation Benchmark.** This research interest revolves around making evaluating LLMs and making those transparent and understandable, particularly in how decisions are made within these systems :
- a) [Adversarialness Evaluation Metric (NAACL 2025) [🔗]],
 - b) [Answer Correctness Evaluation Metric (EMNLP 2024) [🔗]],
 - c) [Correlating VLMs with brain activities (ICLR 2025) [🔗]]
 - d) [LLM Evaluation of metaphor understanding grounded in different cultures (Under Submission)]] e) [Devise a New Adversarialness Evaluation Metric (NAACL 2025) [🔗]],

EDUCATION

Present August 2022	University of Maryland, COLLEGE PARK, PhD in Computer Science and Engineering (CGPA : 3.97/4) Advisor : <i>Prof. Jordan Boyd-Graber</i> Human-in-the-loop AI LLMs and Reasoning Trustworthy systems Generative AI
March 2021 July 2017	Indian Institute of Technology, Kharagpur, , Master of Science (MS) in Computer Science and Engineering Advisors : <i>Prof. Pawan Goyal, Prof. Sudeshna Sarkar</i> CGPA : 9.53/10 (Thesis Submitted : February, 2020) Thesis Title : <i>Predicting similar and interacting entities from structured and unstructured biomedical resources</i>

RESEARCH EXPERIENCE

Present June 2026	Full-Time Research Intern, ADOBE RESEARCH, Speech and Natural Language Processing Advisors : <i>Rajiv Jain</i> > Working on user simulation based on reviews, comments and collaborative decision making
March 2026 September 2025	Student Researcher, GOOGLE RESEARCH, Speech and Natural Language Processing Advisors : <i>Yiwen Song, Yale Song</i> > Working on multi-agent system to incrementally evaluate and generate visual storyboard for target-audience aligned film-making. > Paper Presented at ICML Expo 2026, and under submission [S.2]
August 2025 June 2025	Full-Time Summer Research Intern, ADOBE RESEARCH, Speech and Natural Language Processing Advisors : <i>Dr. Pannaga Shivaswamy , Dr. Aparna Garimella</i> > Working on multi-agent metaevaluator to evaluate the usefulness of a generated video.

May 2025 October 2024	Part-time Student Researcher, MICROSOFT RESEARCH REDMOND, Augmented Learning and Reasoning Advisors : <i>Dr. Jennifer Neville , Dr. Jay Stokes</i> › We extract the 'why,' 'how,' and 'to whom' aspects of personalization from real-world conversational logs and introduce a context-aware preference data augmentation strategy for fine-tuning large language models to cater to diverse user groups.
August 2024 May 2024	Full-Time Summer Research Intern, MICROSOFT RESEARCH REDMOND, Augmented Learning and Reasoning Advisors : <i>Dr. Jennifer Neville , Dr. Jay Stokes</i> › Working on personalization of conversational agents through the study of Copilot Logs for diverse user groups in the Augmented Learning and Reasoning (ALR) Group
September 2023 June 2023	Full-Time Summer Research Intern, ADOBE RESEARCH, Speech and Natural Language Processing Advisors : <i>Dr. Anandhavelu Natarajan , Dr. Aparna Garimella</i> › Working on alignment of document transformation towards diverse persona › Patent Filed and Paper accepted to EACL 2024 [C.3]
July 2022 July 2020	Research Fellow, MICROSOFT RESEARCH INDIA, Speech and Natural Language Processing Advisors : <i>Dr. Kalika Bali, Dr. Monojit Choudhury</i> › Developing an end-to-end multilingual conversational agent which helps the patients understand their medical reports better. › Exploring the gaps between the healthcare chatbots laid out as prototypes in the NLP based literature and the ones actually being deployed to use. › Developing a pipeline for grapheme to phoneme converter for the low-resource Indian languages. › Developing a linguistic annotation schema and annotation interface for the multilingual HIV living youth in Kenya, participating in an online peer support forum and performing sociolinguistic studies on their engagement patterns.
March 2020 April 2017	Project Officer, IIT KHARAGPUR, Computer Science and Engineering Department Advisors : <i>Prof. Sudeshna Sarkar, Prof. Pawan Goyal</i> Industrial Partner : <i>Excelra Knowledge Solutions, Hyderabad</i> › Worked on the main challenges of coreference resolution in mining the information about the biological entities like genes, drugs, diseases from the patents and journals. › Proposed novel methods for Drug-Drug Interaction Prediction and entity linking on medical entities.
June 2020 April 2020	Summer Research Intern, IBM RESEARCH UK, Speech and Natural Language Processing Advisors : <i>Dr. Yufang Hou, Dr. Charles Jochim, Prof. Debasis Ganguly</i> › Developed an end-to-end framework for the construction of NLP-based Knowledge Graph. › Proposed Active-Learning Based methods for enhancing explainability of classification models.
June 2019 April 2019	Summer Research Intern, TCS RESEARCH INDIA, Speech and Natural Language Processing Advisors : <i>Dr. Tirthankar Dasgupta, Dr. Lipika Dey</i> › Worked on implementing attention aware CNN-Bi-LSTM model for inclusion and exclusion criteria classification for the Clinical Trials and proposed a novel weakly-supervised co-training based method which can exploit a large pool of unlabeled criteria sentences to augment the limited supervised training data.

SELECTED PUBLICATIONS

S=IN SUBMISSION/IN PROGRESS, C=CONFERENCE

- [S.4] **Benchmarked Yet Not Measured – Generative AI Should be Evaluated Against Real-World Utility**
Ishani Mondal, Shweta Bhardwaj
(Under Submission) [📄] [[📄]]
- [S.3] **Helping Figures Tell their Story! Paper-Grounded Video Generation Explaining Complex Scientific Figures**
Ishani Mondal, Javad Baghirov, Jordan Boyd-Graber
(Under Submission) [📄] [[📄]]
- [S.2] **CANVAS : Continuity-Aware Narratives for Visual Agentic StoryBoard Generation using a Multi-Agent System**
Ishani Mondal, Yiwen Song, Mihir Parmar, Palash Goyal, Jordan Boyd-Graber, Tomas Pfister, Yale Song
(Under Submission) [📄] [[📄]]
- [S.1] **A Good Talk Doesn't Look Like a Summary, It Teaches You! Measuring Takeaways from Paper-to-Video Talks using Multi-Agent MetaEvaluator**
Ishani Mondal, Aparna Garimella, Ananya Sai, Jordan Boyd-Graber, Pannaga Shivaswamy
(Under Submission) [📄] [[📄]]
- [C.12] **Large language models are effective human annotation assistants, but not good independent annotators**
 Feng Gu, Zongxia Li, Carlos Rafael Colon, Benjamin Evans, Ishani Mondal, Jordan Boyd-Graber
(ACL 2026) [📄] [[📄]]
- [C.11] **NLP for Social Good : A Survey and Outlook of Challenges, Opportunities and Responsible Deployment**
 Antonia Karamolegkou, Angana Borah, Eunjung Cho, Sagnik Ray Choudhury, Martina Galletti, Pranav Gupta, Oana Ignat, Priyanka Kargupta, Neema Kotonya, Hemank Lamba, Sun-Joo Lee, Arushi Mangla, Ishani Mondal, Fatima Zahra Mouda-kir, Deniz Nazar, Poli Nemkova, Dina Pisarevskaya, Naqee Rizwan, Nazanin Sabri, Keenan Samway, Dominik Stamm-bach, Anna Steinberg Schulten, David Tomás, Steven R Wilson, Bowen Yi, Jessica H Zhu, Arkaitz Zubiaga, Anders Sø-gaard, Alexander Fraser, Zhijing Jin, Rada Mihalcea, Joel R. Tetreault, Daryna Dementieva
(EACL 2026) [📄] [[📄]]
- [C.10] **SMART-Editor : A Multi-Agent Framework for Human-Like Design Editing with Structural Integrity**
Ishani Mondal, Meera Bharadwaj, Ayush Roy, Aparna Garimella, Jordan Boyd-Graber
(EACL 2026) [📄] [[📄]]
- [C.9] **Group Preference Alignment : Customized LLM Response Generation from In-Situ Conversations**
Ishani Mondal, Jack W. Stokes, Sujay Kumar Jauhar, Longqi Yang, Mengting Wan, Xiaofeng Xu, Xia Song, Jordan Boyd-Graber, Jennifer Neville
(EMNLP 2025) [📄] [[📄]]
- [C.8] **Correlating instruction-tuning (in multimodal models) with vision-language processing (in the brain)**
 Akshett Rai Jindal, Subba Reddy Oota, Ishani Mondal, Khushbu Pahwa, Satya Sai Srinath Namburi GNVV, Manish Shri-vastava, Maneesh Kumar Singh, Bapi Raju Surampudi, Manish Gupta
(ICLR 2025) [📄] [[📄]]
- [C.7] **Is your benchmark truly adversarial? AdvScore : Evaluating Human-Grounded Adversarialness**
 Yoo Yeon Sung, Maharshi Gor, Eve Fleisig, Ishani Mondal, Jordan Lee Boyd-Graber
(NAACL 2025 (Outstanding Paper Award)) [📄] [[📄]]
- [C.6] **ADAPTIVE IE : Investigating the Complementarity of Human-AI Collaboration to Adaptively Extract Information on-the-fly**
Ishani Mondal, Michelle Yuan, Anandhavelu N, Aparna Garimella, Francis Ferraro, Andrew Blair-Stanek, Benjamin Van Durme, Jordan Boyd-Graber
(COLING 2025 (Oral)) [📄] [[📄]]
- [C.5] **PEDANTS : Cheap but Effective and Interpretable Answer Equivalence**
 Zongxia Li, Ishani Mondal, Yijun Liang, Huy Nghiem, Jordan Boyd-Graber
(EMNLP 2024) [📄] [[📄]]
- [C.4] **SciDoc2Diagrammer-MAF : Towards benchmarking of Scientific Diagram Creation from Documents with Multi-Aspect Refinement**
Ishani Mondal, Zongxia Li, Yufang Hou, Anandhavelu N, Aparna Garimella, Jordan Boyd-Graber
(EMNLP 2024) [📄] [[📄]]
- [C.3] **Presentations by the Humans and For the Humans : Harnessing LLMs for Generating Persona-Aware Slides from Documents**
Ishani Mondal, Shweta S, Anandhavelu N, Aparna Garimella, Sambaran Bandopadhyay, Jordan Boyd-Graber
(EACL (Oral) 2024) [📄] [[📄]]
- [C.2] **End-to-End Construction of NLP Knowledge Graph** [📄][📄]
Ishani Mondal, Yufang Hou, Charles Jochim
Annual Conference of the Association for Computational Linguistics Findings of ACL'21 [Findings of ACL'21]
- [C.1] **BBAEG : Towards BERT-based Biomedical Adversarial Example Generation for Text Classification** [📄][📄]
Ishani Mondal
North American Chapter of the Association for Computational Linguistics NAACL'21 [NAACL'21]

- [C.9] **Super-NaturalInstructions : Generalization via Declarative Instructions on 1600+ NLP Tasks**
Yizhong Wang,..., Ishani Mondal....
EMNLP 2022 [📄] [📄]
- [C.8] **Are the predictions and explanations provided by ChatGPT robust under Adversarial Attack?**
Ishani Mondal, Abhilasha Sancheti
(ICLR Tiny 2024) [📄] [📄]
- [C.7] **Intent Identification and Entity Extraction for Healthcare Queries in Indic Languages**
Ankan Mullick, Ishani Mondal, Sourjyadip Ray, Raghav R, G Sai Chaitanya and Pawan Goyal
EACL Findings 2023 [📄] [📄]
- [C.6] **Explaining (Sarcastic) Utterances to Enhance Affect Understanding in Multimodal Dialogues**
Shivani Kumar, Ishani Mondal, Md Shad Akhtar, Tanmoy Chakraborty
AAAI 2023 [📄] [📄]
- [C.5] **Global Readiness of Language Technology for Healthcare : What would it Take to Combat the Next Pandemic?** [📄]
Ishani Mondal, Kabir Ahuja, Mohit Jain, Jacki O'Neil, Kalika Bali, Monojit Choudhury
COLING 2022 [COLING 2022]
- [C.4] **Understanding Sociolinguistic and Pragmatic Aspects of Peer Support in Multilingual Healthcare Conversations**
Ishani Mondal, Mohit Jain, Kalika Bali, Monojit Choudhury, Jacki O'Neil
LREC 2022 [📄] [📄]
- [C.3] **Multi-Objective Few-shot Learning for Fair Classification** [📄][📄]
Ishani Mondal, Procheta Sen, Debasis Ganguly
ACM International Conference on Information and Knowledge Management CIKM'21 [CIKM'21]
- [C.2] **ALEX : Active Learning based Enhancement of a Classification Model's EXplainability** [📄][📄]
Ishani Mondal, Debasis Ganguly
ACM International Conference on Information and Knowledge Management CIKM'20 [CIKM'20]
- [C.1] **Drug-Drug Interactions Prediction Based on Drug Embedding and Graph Auto-Encoder** [📄]
Ishani Mondal, Sukannya Purkayastha, Sudeshna Sarkar, Pawan Goyal, Jitesh K Pillai
IEEE 19th International Conference on Bioinformatics and Bioengineering IEEE BIBE'19 [IEEE BIBE'19]

WORKSHOP/SYMPOSIUM PUBLICATIONS

S=IN SUBMISSION/IN PROGRESS

- [W.8] **MunTTS : A Text-to-Speech System for Mundari** [📄]
Varun Gumma, Rishav Hada, Aditya Yadavalli, Pamir Gogoi, Ishani Mondal, Vivek Seshadri, Kalika Bali
The Use of Computational Methods in the Study of Endangered Languages ComputEL-7@EACL'24 [ComputEL-7@EACL'24]
- [W.7] **A Linguistic Annotation Framework to Study Interactions in Multilingual Healthcare Conversational Forums** [📄]
Ishani Mondal, Kalika Bali, Mohit Jain, Monojit Choudhury, Ashish Sharma, Evans Gitau, Jacki O'Neill, Kagonya Awori, Sarah Gitau
15th Linguistic Annotation Workshop (LAW) and 3rd Designing Meaning Representations (DMR) Workshop LAW-DMR@EMNLP'21 [LAW-DMR@EMNLP'21]
- [W.6] **BERTKG-DDI : Towards Incorporating Entity-specific Knowledge Graph Information in Predicting Drug-Drug Interactions** [📄]
Ishani Mondal
Workshop on Scientific Document Understanding (SDU) SDU@AAAI'21 [SDU@AAAI'21]
- [W.5] **BERTChem-DDI : Improved Drug-Drug Interaction Prediction from text using Chemical Structure Information** [📄]
Ishani Mondal
Knowledgeable NLP : the First Workshop on Integrating Structured Knowledge and Neural Networks for NLP KnowledgeableNLP@AAACL-IJCNLP'20 [KnowledgeableNLP@AAACL-IJCNLP'20]
- [W.4] **Extracting Semantic Aspects for Structured Representation of Clinical Trial Eligibility Criteria** [📄]
Ishani Mondal, Tirthankar Dasgupta, Abir Naskar, Sudeshna Jana, Lipika Dey.
3rd Clinical Natural Language Processing Workshop ClinicalNLP@EMNLP'20 [ClinicalNLP@EMNLP'20]

- [W.3] **Automatic Segregation and Classification of Inclusion and Exclusion Criteria of Clinical Trials to improve Patient Eligibility Matching** [🔗]
 Tirthankar Dasgupta, [Ishani Mondal](#), Abir Naskar, Sudeshna Jana, Lipika Dey
Explainable AI in Healthcare and Medicine HealthIntelligence@AAAI'20 [HealthIntelligence@AAAI'20]
- [W.1] **Medical Entity Linking using Triplet Network** [🔗]
[Ishani Mondal](#), Sukannya Purkayastha, Sudeshna Sarkar, Pawan Goyal, Jitesh Pillai, Amitava Bhattacharyya, Mahanandeeswar Gattu
2nd Clinical Natural Language Processing Workshop ClinicalNLP@NAACL'19 [ClinicalNLP@NAACL'19]

ACADEMIC SERVICE

Program Committee	CIKM 2025, CIKM 2026
Reviewer	ARR March 2026, COLM 2026, CIKM 2025, EMNLP 2025, ARR January 2026, ARR May 2024, PeerJ EMNLP'23, ACL'23, ClinicalNLP@EMNLP'23, BMC Bioinformatics 2021, MLH4H@NeurIPS'21
Sub-Reviewer	ACL'21, NAACL'21, WSDM'21, LREC'20, CODS-COMAD'20, Code-Switching Workshop@EMNLP'21, RAI Workshop@KDD'21
Volunteer	GHCI'19, IRISS'19, ISCLS'19

HONOURS AND AWARDS

Outstanding Paper Award from NAACL 2025
 Dean's Fellowship from University of Maryland (2022-2024)
 Batch Topper by Computer Science Department, Institute of Engineering and Management in 2015
 Merit Scholarship awarded in 2011 for being consistent topper in high-school
 Grace-Hopper India Scholarship'19 [🔗] For attending and presenting poster in 2019
 Infosys Women Travel Grant [🔗] For attending and presenting paper at NAACL in 2019

TEACHING AND LEADERSHIP ROLES

SNLP Reading Group, MSR India <i>Organizer</i>	JUL'21 - PRESENT
> Organize a weekly lab-wide Reading Group focused on research taking place in the area of Speech and Natural Language Processing (SNLP). We read recent, classical papers as well as arrange for external talks in related areas.	
Technology for Emerging Market Reading Group, MSR India <i>Organizer</i>	DEC'21 - PRESENT
> Organize a weekly lab-wide Reading Group focused on research taking place in the area of User-centered Technologies. We read classical papers and arrange for invited talks.	
Introduction to Speech and Natural Language Processing (CS60057) <i>Lead Teaching Assistant</i>	JULY'19 - DEC'19
> Course Instructor : Prof. Sudeshna Sarkar > Responsibilities included evaluating labs, and helping students with the coursework, lab assignments and projects.	
Introduction to Machine Learning by NPTEL <i>Teaching Assistant</i>	JULY'19 - DEC'19
> Course Instructor : Prof. Sudeshna Sarkar > Responsibilities included preparation of question papers and assignments for the students.	
Introduction to Deep Learning (CS60010) <i>Teaching Assistant</i>	JAN'19 - MAY'19
> Course Instructor : Prof. Sudeshna Sarkar > Responsibilities included evaluating labs, and helping students with the coursework and lab assignments.	
Introduction to Speech and Natural Language Processing (CS60057) <i>Teaching Assistant</i>	JULY'18 - DEC'18
> Course Instructor : Prof. Sudeshna Sarkar > Responsibilities included evaluating labs, and helping students with the coursework, lab assignments and projects.	
Introduction to Programming and Data Structure (CS10001) <i>Teaching Assistant</i>	JAN'18 - MAY'18
> Course Instructor : Prof. Aritra Hazra > Responsibilities included evaluating labs, and helping students with the coursework and lab assignments.	

TALKS AND PRESENTATIONS

“Multi-objective Few-shot Learning for Fair Classification”[📄](November 2021 (Remote))

“End to End Construction of Scientific Knowledge Graph”[📄](July 2021 (Remote))

“BBAEG : Towards BERT-based Biomedical Adversarial Example Generation for Text Classification”[📄](June 2021 (Remote))

“ALEX : Active learning based enhancement of a classification model’s EXplainability”[📄](November 2020 (Remote))

TECHNICAL SKILLS

Languages	Python, C, Java , MATLAB, Bash Scripts
Frameworks	PyTorch, Keras, TensorFlow, NumPy, Pandas, Scikit-Learn, Scipy, NLTK, Spacy, Scispacy
DevOps and Version Control	Microsoft Azure, Amazon Web Services, Google Cloud, Git, SVN
Relevant CourseWork	Machine Learning, Deep Learning, Speech and Natural Language Processing, Information Retrieval, Algorithms, Programming and Data Structures, Object Oriented Programming, Discrete Mathematics

REFERENCES

- Prof. Jordan Boyd-Graber Associate Professor, Computer Science Engineering Department, University of Maryland [📄]
- Prof. Hal Daume III Associate Professor, University of Maryland and Senior Principal Researcher at Microsoft Research NYC [📄]
- Prof. Jennifer Neville Principal Research Manager at Microsoft Redmond and Professor at Purdue University [📄]
- Prof. Monojit Choudhury Professor of Natural Language Processing, MBZUAI [📄]
- Prof. Yufang Hou Prof of NLP at IT:U - Interdisciplinary Transformation University Austria [📄]
- Dr. Kalika Bali Principal Researcher, Microsoft Research, India [📄]
- Dr. Aparna Garimella Research Scientist, Adobe Research, India [📄]
- Prof. Pawan Goyal Associate Professor, Computer Science Engineering Department, IIT Kharagpur [📄]
- Prof. Sudeshna Sarkar Professor, Computer Science Engineering Department, IIT Kharagpur [📄]
- Prof. Abhay Karandikar Professor and Director, Department of Electrical Engineering, IIT Kanpur [📄]
- Prof. Debasis Ganguly Assistant Professor, University of Glasgow, UK [📄]